

Data Preparation

Objective

The objective of this step is to prepare the electricity consumption dataset for visualization in Tableau. Since the dataset was already pre-cleaned, the focus is on reviewing, validating, and formatting the data to ensure it is suitable for analysis.

Dataset Review

The imported dataset was reviewed to understand its structure, available fields, and overall quality. All columns were examined to verify that the data was complete and suitable for visualization.

Data Validation

The dataset was checked to confirm that there were no missing values, duplicate records, or inconsistencies. Since the dataset had already been cleaned, no additional data cleaning was required.

Data Type Verification

Each field was verified to ensure that it had the correct data type. Date fields were assigned the Date data type, numerical fields such as electricity consumption were verified as numeric values, and categorical fields such as state and region were maintained as string values.

Field Formatting

Field names were reviewed for clarity and consistency. Date formatting was also verified to ensure that Tableau could correctly interpret the data for time-based analysis.

Data Structuring

The dataset structure was confirmed to support the project objectives. No additional filtering or restructuring was required because the available data already matched the analysis requirements.

Calculated Fields

No calculated fields were created during this stage. Any additional calculations required for analysis will be developed during the dashboard creation phase.

Result

The dataset was successfully prepared and validated for visualization in Tableau. All fields were correctly formatted, data quality was verified, and the dataset was confirmed to be ready for creating worksheets, dashboards, and analytical insights.